

A fixed-point-free condition- (L) map on a weakly compact convex subset of c_0

Solution packet for arXiv:1209.5368, Remark 4.8

13 August 2026

Result. There is a weakly compact convex set $K \subset c_0$ and a Borel self-map $T : K \rightarrow K$ with no fixed point which satisfies condition (L) , hence also property $(*)$. Moreover c_0 is NUNC and $M(c_0) = MW(c_0) = 2$. Thus the proposed extension in Remark 4.8 is false under each of these three hypotheses. The uniformly nonsquare subcase is not decided here.

1. Source question and answer

Betiuk-Pilarska and Wiśnicki prove fixed-point theorems for maps satisfying Suzuki's condition (C) (or continuous maps satisfying (C_λ)) on weakly compact convex sets under the NUNC, $M(X) > 1$, and $MW(X) > 1$ hypotheses. Their final Remark 4.8 asks whether the results remain true for maps satisfying (L) or $(*)$ [1]. The following is a crop of page 14 of the official arXiv PDF; the geometric lead-in and the exact question are retained.

Recall that a Banach space X is uniformly nonsquare if

$$J(X) = \sup_{x,y \in S_X} \min \{ \|x + y\|, \|x - y\| \} < 2.$$

In [9], a characterization of reflexive Banach spaces with $MW(X) > 1$ is given. In particular (see [9] Corollary 5.1]), all uniformly nonsquare Banach spaces fulfill this condition. Thus we obtain the following corollary which answers Question 1 in [3].

Corollary 4.7. *Let C be a nonempty convex weakly compact subset of a uniformly nonsquare Banach space. Then every mapping $T : C \rightarrow C$ which satisfies condition (C) and every continuous mapping $T : C \rightarrow C$ which satisfies condition (C_λ) for some $\lambda \in (0, 1)$, has a fixed point.*

Remark 4.8. It is not known whether our results are valid for mappings satisfying property (L) or $(*)$.

Acknowledgement. The authors thank Mariusz Szczepanik for helpful discussions and drawing their attention to Theorem 9 in [18].

Official source evidence: arXiv:1209.5368, p. 14.

Recall only the two clauses of (L) that will be used. A map $T : C \rightarrow C$ satisfies (L) when (i) every nonempty closed convex invariant subset contains an approximate fixed-point sequence (AFPS), and (ii), for every AFPS (x_j) and $x \in C$,

$$\limsup_j \|x_j - Tx\| \leq \limsup_j \|x_j - x\|.$$

The source observes that (L) implies $(*)$.

2. Proof intuition

The construction needs an orbit with two apparently conflicting features. It must escape weakly so that its closed convex hull is weakly compact, yet successive orbit points must become arbitrarily close so that every invariant subset acquires an AFPS. Wide triangular “tents” moving to the right in c_0 do both: their supports escape every coordinate, while their centers move much less than their widths.

For an arbitrary point of the convex hull, send it to the successor of its nearest tent. Any AFPS must then be close to tents whose indices tend to infinity. Such a sequence has asymptotic distance exactly one from *every* point of the convex hull. Condition (L) becomes the equality $1 \leq 1$, whereas a fixed point is impossible because the map always moves one step forward.

3. The moving tents

Put, for $n \geq 1$ and $k \in \mathbb{N}$,

$$c_n = n^4, \quad w_n = n^{7/2}, \quad u_n(k) = \max \left\{ 0, 1 - \frac{|k - c_n|}{w_n} \right\}.$$

Thus $u_n \in c_0$, $0 \leq u_n(k) \leq 1$, and $\|u_n\|_\infty = 1$.

Lemma 1. *The sequence (u_n) is weakly null. If $d_n = \|u_{n+1} - u_n\|_\infty$, then $d_n > 0$ for every n and $d_n \rightarrow 0$.*

Proof. The left endpoint of the support is $c_n - w_n = n^{7/2}(\sqrt{n} - 1) \rightarrow \infty$. For $a = (a_k) \in (c_0)^* = \ell_1$,

$$|\langle a, u_n \rangle| \leq \sum_{k \geq c_n - w_n} |a_k| \rightarrow 0,$$

so $u_n \rightarrow 0$.

For the step estimate, let $\phi_{c,w}(t) = \max\{0, 1 - |t - c|/w\}$ on $\mathbb{R}$. If $w' \geq w$, widening a tent and then translating its center gives

$$\|\phi_{c',w'} - \phi_{c,w}\|_\infty \leq \frac{|c' - c|}{w'} + 1 - \frac{w}{w'}.$$

(The first term is the Lipschitz translation bound; the second is the maximum change caused by widening.) Restriction to integer coordinates and $c_n = n^4, w_n = n^{7/2}$ give

$$d_n \leq \frac{(n+1)^4 - n^4}{(n+1)^{7/2}} + 1 - \left(\frac{n}{n+1} \right)^{7/2} \rightarrow 0.$$

Finally u_n and u_{n+1} have different unique peak coordinates, so $d_n > 0$. □

Let

$$K = \overline{\text{co}}^{\|\cdot\|} \{0, u_1, u_2, \dots\} \subset c_0.$$

The set $\{0, u_1, u_2, \dots\}$ is weakly compact by Lemma 1; the closed-convex-hull theorem therefore makes K weakly compact. It is nonempty and convex, and all coordinates of every $z \in K$ lie in $[0, 1]$.

Lemma 2. *For every $z \in K$, $\|u_n - z\|_\infty \rightarrow 1$. Consequently every $x \in K$ has a least-index nearest orbit point.*

Proof. Since $c_n \rightarrow \infty$ and $z \in c_0$, $z(c_n) \rightarrow 0$. Positivity gives

$$1 - z(c_n) \leq \|u_n - z\|_\infty \leq 1,$$

which proves the limit. Put $r_n = \|x - u_n\|_\infty \leq 1$. If $\inf_n r_n < 1$, the tail $r_n \rightarrow 1$ leaves only a finite minimization. If $\inf_n r_n = 1$, then $r_n = 1$ for every n . In both cases a minimum is attained, and the least minimizing index exists. □

Define

$$\nu(x) = \min_{n \geq 1} \argmin \|x - u_n\|_\infty, \quad T(x) = u_{\nu(x)+1}.$$

4. Classification of all approximate fixed points

Lemma 3. *If (x_j) is an AFPS for T in K , then for every $z \in K$,*

$$\|x_j - z\|_\infty \longrightarrow 1.$$

Proof. Write $n_j = \nu(x_j)$ and $\delta_j = \|Tx_j - x_j\|_\infty = \|u_{n_j+1} - x_j\|_\infty \rightarrow 0$. Minimality of u_{n_j} yields

$$\|x_j - u_{n_j}\|_\infty \leq \delta_j, \quad d_{n_j} \leq \|u_{n_j} - x_j\|_\infty + \|x_j - u_{n_j+1}\|_\infty \leq 2\delta_j.$$

Because every d_n is strictly positive, the last convergence forces $n_j \rightarrow \infty$: otherwise some fixed index would recur along a subsequence. Lemma 2 and $\|x_j - u_{n_j}\|_\infty \rightarrow 0$ now give

$$|\|x_j - z\|_\infty - \|u_{n_j} - z\|_\infty| \leq \|x_j - u_{n_j}\|_\infty \longrightarrow 0,$$

and the claimed limit follows. $\square$

Theorem 1. *The self-map $T : K \rightarrow K$ has no fixed point and satisfies condition (L). It also satisfies property (*) directly.*

Proof. First, $\nu(u_m) = m$. If $Tx = x$, then $x = u_m$ with $m = \nu(x) + 1$, while $\nu(x) = \nu(u_m) = m$, a contradiction.

Let $D \subset K$ be nonempty, closed, convex, and T -invariant. Choose $x \in D$. Then $Tx = u_m \in D$ for some m , and $T(u_r) = u_{r+1}$ puts the whole tail $(u_r)_{r \geq m}$ in D . This is an AFPS by Lemma 1, proving clause (i) of (L).

For any AFPS (x_j) and any $x \in K$, Lemma 3, applied first to $Tx \in K$ and then to $x \in K$, gives

$$\limsup_j \|x_j - Tx\|_\infty = 1 = \limsup_j \|x_j - x\|_\infty.$$

Thus clause (ii) holds with equality.

For completeness, take invariant closed convex $D, E \subset K$ and an AFPS in D . Every point of E has asymptotic distance one by Lemma 3; hence its asymptotic center relative to E is all of E , which is invariant. Together with the preceding tail argument this verifies (*) without invoking (L) $\Rightarrow$ (*). $\square$

The map is Borel. Indeed, for continuous $r_n(x) = \|x - u_n\|_\infty$,

$$\{\nu = n\} = \bigcap_{m \geq 1} \{r_n \leq r_m\} \cap \bigcap_{m < n} \{r_n < r_m\}.$$

Thus no nonmeasurable selection is hidden in the construction.

5. Why c_0 meets the paper's hypotheses

Proposition 1. *For every $a \geq 0$,*

$$R(a, c_0) = RW(a, c_0) = \max\{1, a\}.$$

Consequently $M(c_0) = MW(c_0) = 2$, and c_0 is NUNC.

Proof. Let (y_n) be weakly null in B_{c_0} and $x \in c_0$. Given $\eta > 0$, choose a finite set F such that $|x(k)| < \eta$ off F . Coordinatewise convergence makes $\max_{k \in F} |y_n(k)| < \eta$ eventually, while off F we have $|y_n(k) \pm x(k)| \leq 1 + \eta$. Hence

$$\limsup_n \|y_n \pm x\|_\infty \leq \max\{1, \|x\|_\infty\}.$$

For $x = ae_1$ and $y_n = e_{n+1}$, both signs attain $\max\{1, a\}$, and $D[(y_n)] = 1$. This proves both modulus formulas, including the asymptotic-diameter restriction in R . Therefore

$$M(c_0) = MW(c_0) = \sup_{a \geq 0} \frac{1+a}{\max\{1, a\}} = 2.$$

For NUNC, fix $x \in S_{c_0}$, a weakly null $(y_n) \subset S_{c_0}$, and $0 < t \leq 1$. The norm of x is attained at some coordinate k_0 , so $\liminf_n \|x + ty_n\|_\infty \geq 1$ because $y_n(k_0) \rightarrow 0$. Repeating the finite/tail estimate above gives the reverse limsup bound; hence $\|x + ty_n\|_\infty \rightarrow 1$. In the source notation this says $b(t, x) = 0$ for every $x \in S_{c_0}$ and $0 < t \leq 1$. For each $\varepsilon > 0$, choose (say) $t = 1/2$; then $b(t, x) = 0 \leq \varepsilon t$ uniformly in x , which is precisely the second NUNC alternative. $\square$

Combining Theorem 1 and Proposition 1 answers Remark 4.8 negatively for NUNC, $M(X) > 1$, and $MW(X) > 1$. Since c_0 is not uniformly nonsquare (take $x = e_1 + e_2$, $y = e_1 - e_2$), the uniformly nonsquare specialization is not resolved by this packet.

6. Reproducibility and literature scope

The included script `code/verify_tents.py` exhaustively checks the first sampled tent steps, the analytic step bound, large-index decay using the piecewise-linear knots, nearest-point behavior on orbit points and random finite convex combinations, and the inequalities used in the AFPS classification. Run it with:

```
python3 code/verify_tents.py
```

The expected first line is `moving-tent checks: PASS`. These finite checks are diagnostic; all infinite-dimensional claims are proved above.

Exact-ID and core-phrase searches of the run indexes found no duplicate. A bounded literature search through 2026-08-13 found general fixed-point-free (L) examples, but no later answer with the simultaneous weak compactness and c_0 /NUNC/ M geometry used here. The claimed result is the explicit counterexample above, not a claim about the uniformly nonsquare subcase or every conclusion in the source paper.

References

- [1] A. Betiuk-Pilarska and A. Wiśnicki, *On the Suzuki nonexpansive-type mappings*, arXiv:1209.5368 (2012), especially Definitions 1.3–1.4 and Remark 4.8, <https://arxiv.org/abs/1209.5368>.